CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Memory Organization

Lecture No.- 02



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Recap of Previous Lecture



Topic Memory Hierarchy

Topic Memory Presentation

Topics to be Covered



- **Topic** Memory Address Decoder
- **Topic** Main Memory
- **Topic** RAM & ROM Chip
- **Topic** Multiple Chips in a Memory

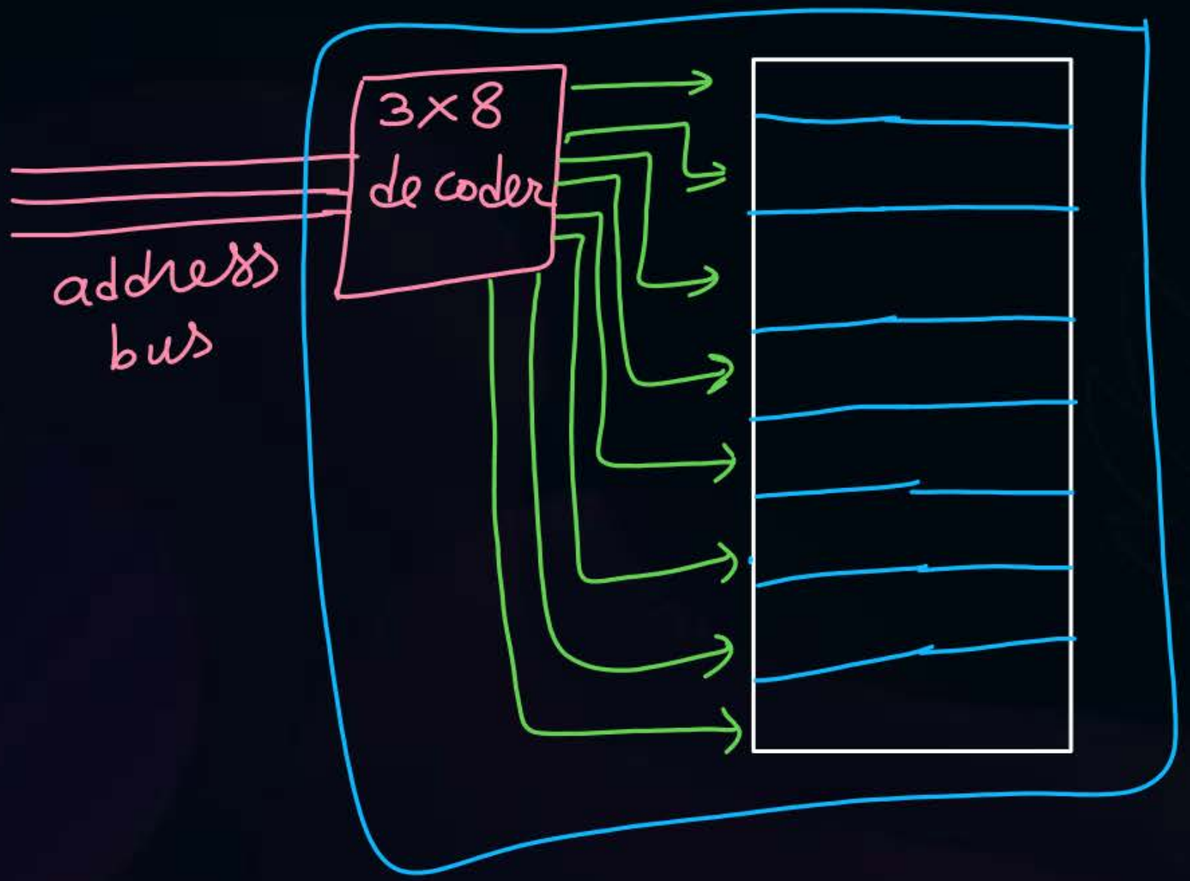


Topic : Memory Address Decoder

ex :-

no. of cells = 8

add. = 3 bits



[NAT]

#Q. Consider a memory of size 2K × 8-bits. What is the size of decoder needed to access the cells of the memory uniquely?

no. of cells = 2k

add. = 11 bits

---

decoder ⇒ $11 \times 2k$

or

11×2^{11}

or

11×2048

[NAT]

GATE-PYQ Ans = 1034

#Q. If there are m input lines n output lines for a decoder that is used to uniquely address a byte addressable 1 KB RAM, then the minimum value of m + n is 1034 ?

1k x 1B

no. of cells = 1k = 1024

add. = 10 bits

decode ⇒ 10 x 1024

m = 10

n = 1024

m + n = 1034



Topic : Main Memory (Primary)

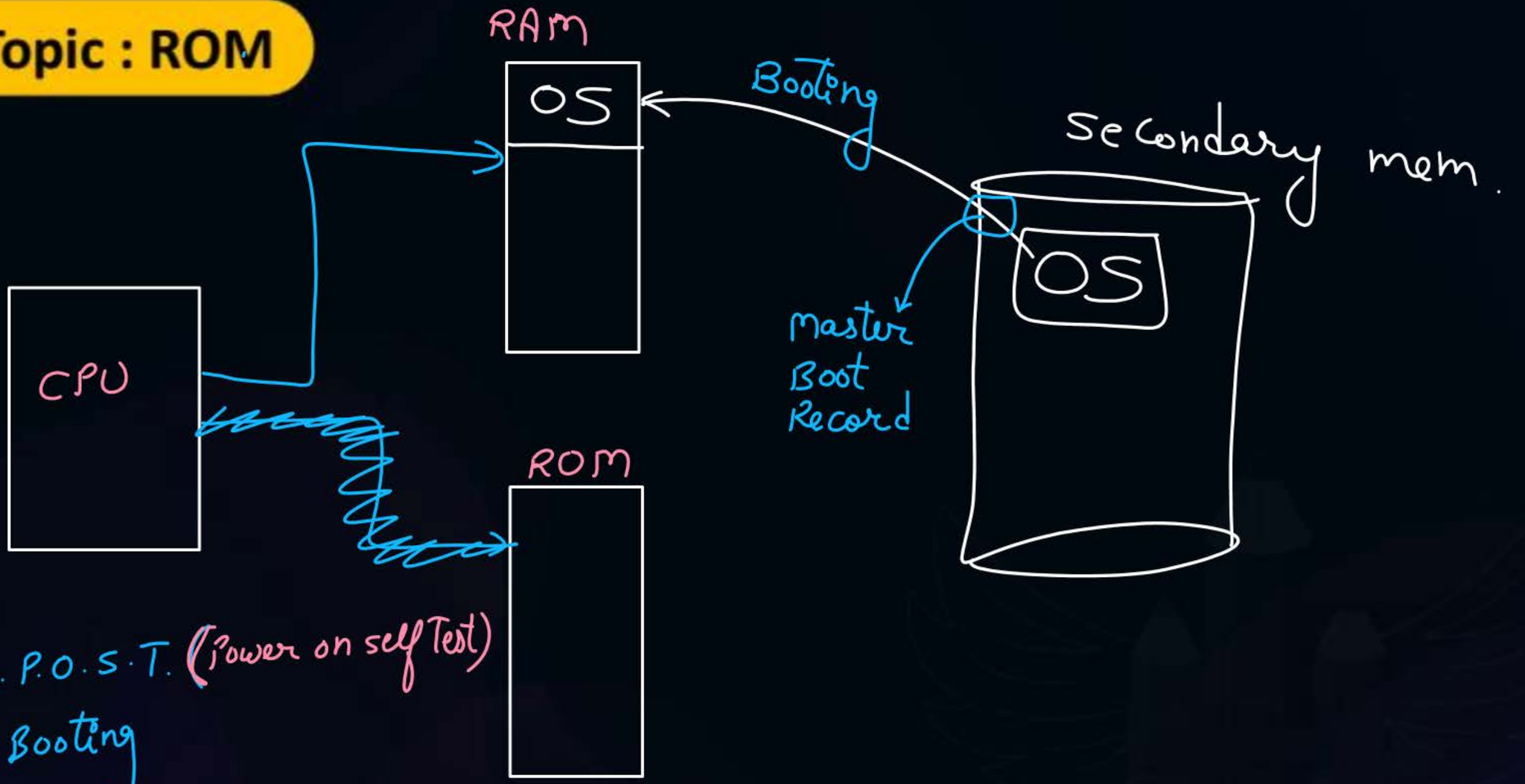
which is used to store current running programs and their data.

Type :-

- RAM ⇒ Volatile.
- ROM ⇒ Non-Volatile



Topic : ROM



1. P.O.S.T. (Power on self Test)
2. Booting



Topic : RAM

↓

main mem.

Types :-

- static RAM (SRAM)
- Dynamic RAM (DRAM)



Topic : Types of RAM

Static	Dynamic
1. Implemented using flip-flops	1. Implemented using capacitors
2. No refresh required	2. Periodic refresh is required
3. Faster Read/Write	3. Slow Read/Write
4. Used for Cache	4. Used for main memory
5. Expensive	5. Less Expensive
6. Low Idle power consumption	6. High Idle power consumption
7. High operational power consumption	7. Low operational power consumption

stores content in form of electric charges

↳ recharge

[NAT]

#Q. Consider 2 4-bits unsigned values A and B. What will be the maximum size of result for:

1. Addition of A and B ⇒ 5 bits | (n+1) bits
2. Multiplication of A and B ⇒ 8 bits | 2n bits

1111 = 15
1111 = 15
(30)$_2$

⇓

5 bits

15
*15
(225)$_{10}$ ⇒ 8 bits

[MCQ]

GATE- 1 PYQ

#Q. The amount of ROM needed to store the table for multiplication of two 4-bit unsigned integer is?

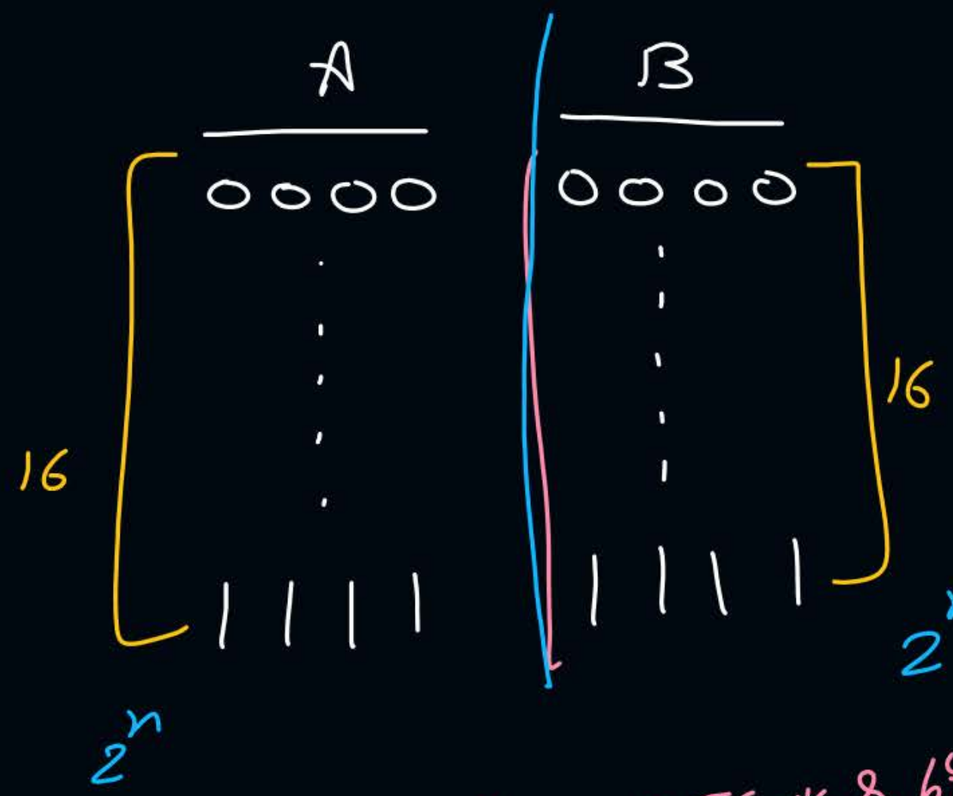
multiplication of each possible value of 2 inputs

A 64 bits

B 128 bits

C 1K bits

D ✓ 2K bits

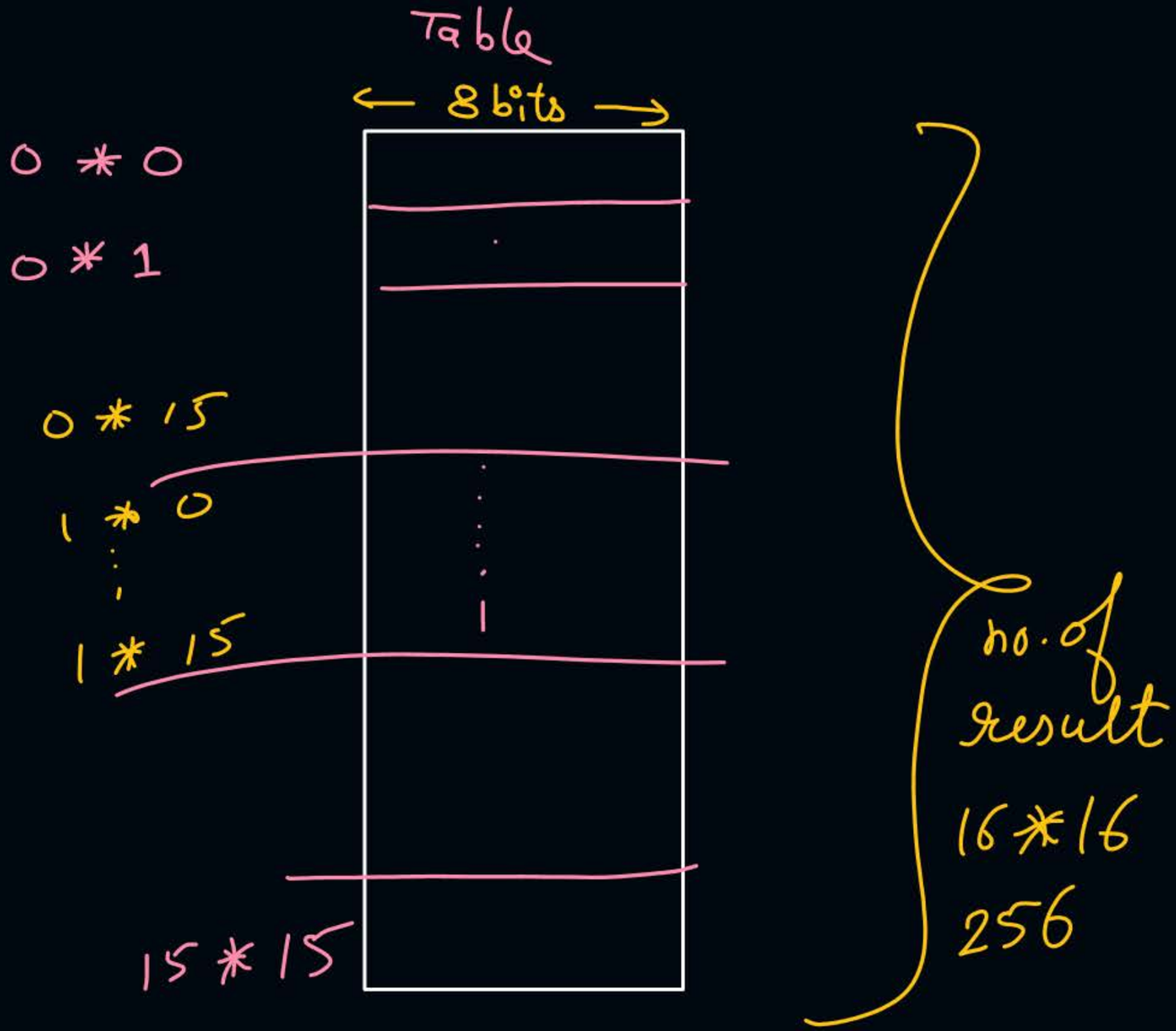


mem size = 256 * 8 bits

$= 2^8 * 2^3$ bits

$= 2^{11}$ bits

= 2K bits



for n bits

multiplication table

$2^{2n} \times 2n$ bits

Addition Table

$2^{2n} \times (n+1)$ bits



Topic : RAM Chip

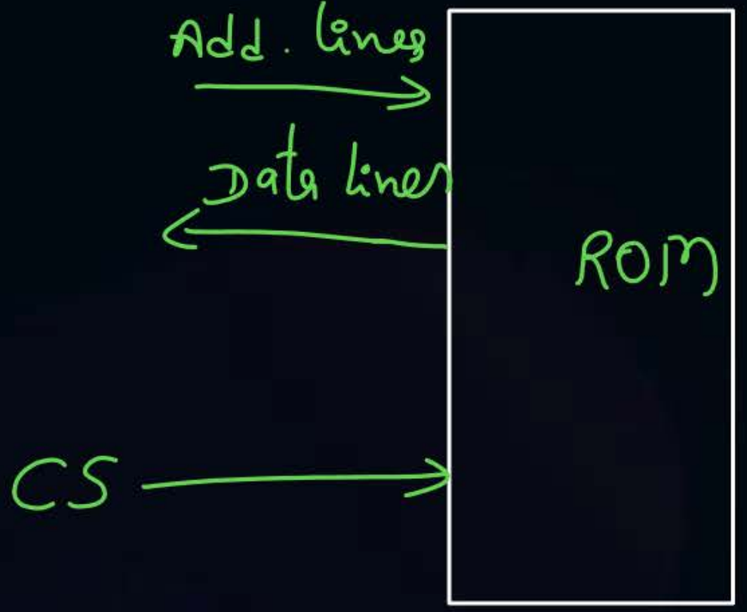


CS	Read	write	operation
0	X	X	No operation
1	0	0	No operation
1	0	1	write
1	1	X	Read

X ⇒ don't care
(0 or 1 we do not care)



Topic : ROM Chip



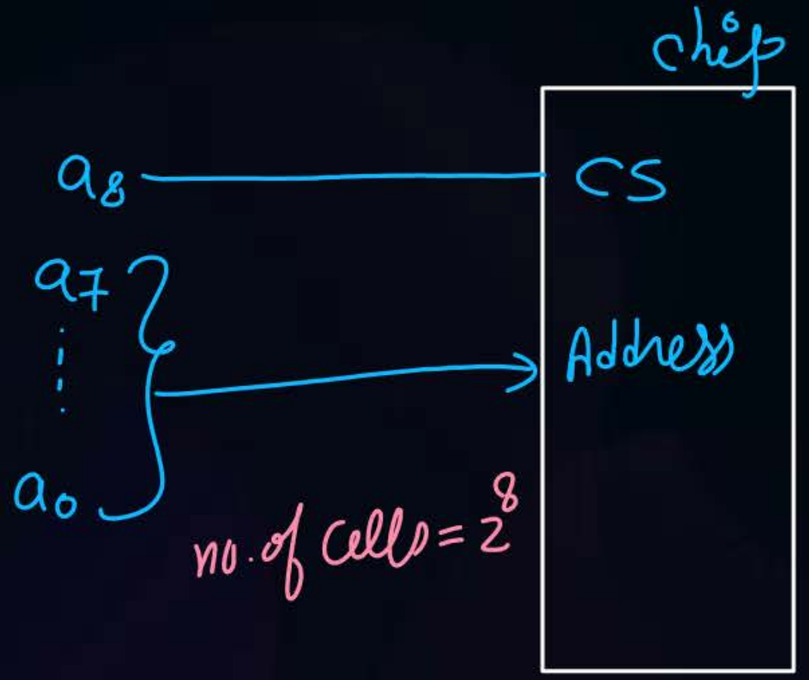
CS	
0	No operation
1	Read



Topic : Chip Select

9-bits add. CPU generated $\Rightarrow$ 8 bits used as address
1 bit used as C.S.

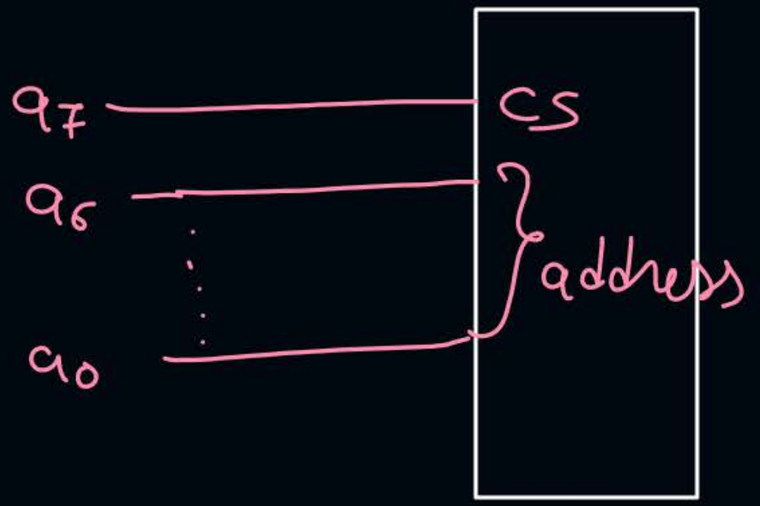
$a_8\ a_7 \cdots a_0$



a_8	a_7	a_6	----	a_0
1	0	0	-------	0
1	0	0	-------	1
1	1	1	-------	1

address to access given chip

ex:- 8 bits address $a_7 \cdots a_0$



address range

1 0 0 0 0 0 0 0

1 1 1 1 1 1 1 1

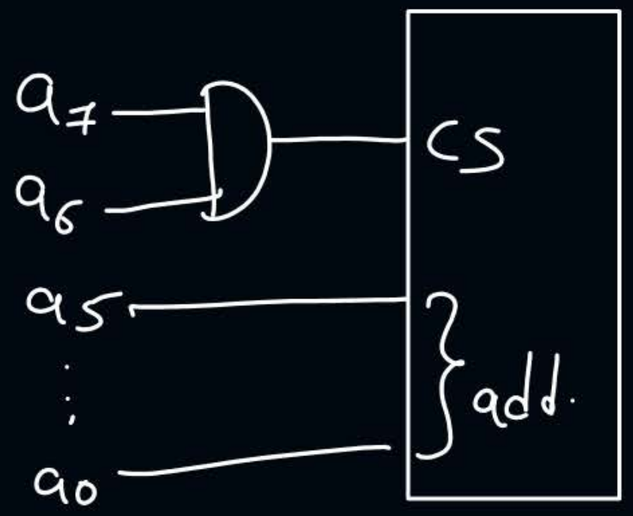
In hexadecimal

80

FF

Ex:-

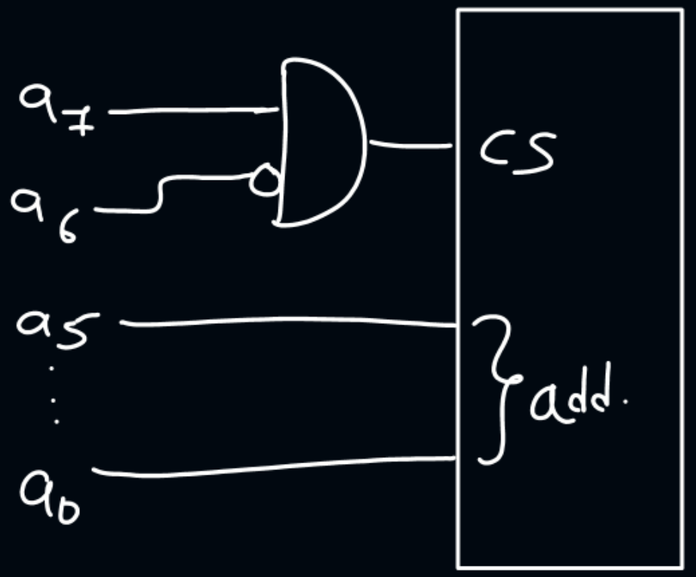
8-bits add => $a_7 - - - a_0$



add. range

								Hexa
1	1	0	0	0	0	0	0	C0
1	1	1	1	1	1	1	1	FF

ex:- 8 bits address



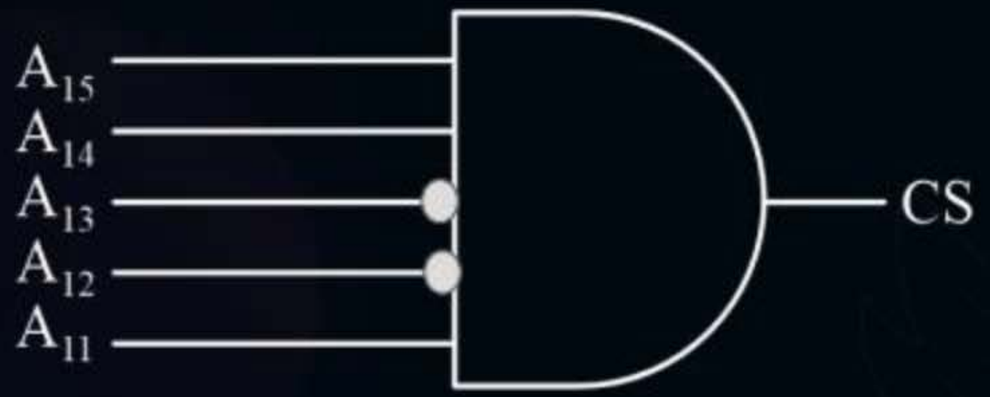
add. range :-

a_7	a_6	...	a_0	(Hexa)
1	0	0 ...	0	80
1	0	1 1 ...	1	BF

[MCQ]

GATE- PYQ

#Q. The chip select logic for a certain DRAM chip in a memory system design is shown below. Assume that the memory system has 16 address lines denoted by A_{15} to A_0. What is the range of address (in hexadecimal) of the memory system that can get enabled by the chip select (CS) signal?



add. range :-

1 1 0 0 | 0 0 0 0 ... 0

1 1 0 0 | 1 1 1 1 ... 1

C800 - CFFF

(A) ✓ C800 to CFFF

(B) CA00 to CAFF

(C) C800 to C8FF

(D) DA00 to DFFF



Topic : Multiple Chips in Single Memory System

Total mem. capacity = no. of chips * 1 chip capacity

$$\text{no. of chips} = \frac{\text{Total capacity}}{\text{1 chip capacity}}$$

[NAT]

Ans = 16

#Q. How many 64 bytes RAM chips are needed to provide a memory capacity of 1Kbytes?

$$= \frac{1\,KB}{64\,B}$$

$$= \frac{2^{10}}{2^6}$$

$$= 2^4$$

$$= 16$$

[NAT]

#Q. Total memory capacity is __8__ Mbytes, if we use 16 chips of size 512Kbytes each?

$= 16 * 512\,KB$

$= 2^4 * 2^{19}\ B$

$= 2^{23}\ B$

$= 8\ mB$



2 mins Summary

Topic Memory Address Decoder

Topic Main Memory

Topic RAM & ROM Chip

Topic Multiple Chips in a Memory



Happy Learning

THANK - YOU